59 => greedy
knapsack

Lec_60

Q. **Job Sequencing Problem (with deadline)**

Problem Definition:

Given an array of jobs where every job has a deadline and associated profit if the job is finished before the deadline. It is also given that every job takes a single unit of time, so the minimum possible deadline for any job is 1. Maximize the total profit if only one job can be scheduled at a time.

1 day

1 JS

work

1 - 30

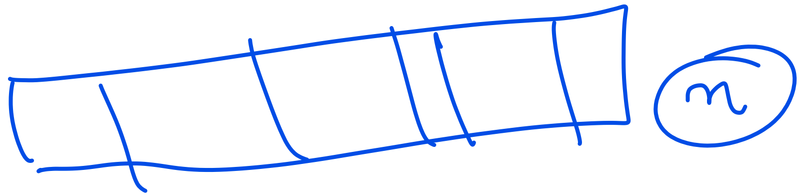
Time 1 Month

J_1 J_2 J_3 . . . J_n

deadline 19 12 5 3 2

pr 26 30 25 200

In computer Science



Conclusion:

- Single CPU Available ⇒ In 1 day only one work
- Arrival time of each job is same ⇒ work is generate at 1st date
- No Premotion → In 1 day only one Job should finished
- Running time of each job is 1-unit

⇓

one day is sufficient to finish one unit for below data, using Job seque algo find i) max^m profit ii) which Job has been left out

Ex1:

Jobs:	J1	J2	J3	J4	J5	J6	J7
Profit:	100	80	70	55	90	150	39
Deadline:	5	3	5	2	3	1	3

(work)

(profit)

(date)

Max^m deadline is = 5

J6	J2	J5	J3	J1
150	80	90	70	100
1	2	3	4	5

i) ⟹ 150 + 80 + 90 + 70 + 100

ii) J4, J7

algorithm Job: J_1 J_2 J_3 J_4 - - - J_n

deadline: d_1 d_2 d_3 d_4 - - - d_n

profit: p_1 p_2 p_3 p_4 - - - p_n

$\Rightarrow$ sort the Job on the basis of profit in decreasing order

Ex2:

Jobs:	J1	J2	J3	J4	J5	J6	J7	J8	J9
Profit:	75	25	85	95	45	35	100	150	65
Deadline:	5	7	5	3	4	5	3	1	5

Max^m deadline = 7

Step 1 Sort the Jobs in decreasing order

Jobs	J8	J7	J4	J3	J1	~~J9~~	~~J5~~	~~J6~~	J2
profit	150	100	95	85	75	~~65~~	~~45~~	~~35~~	25
Deadline	1	3	3	5	5	5	4	5	7

Step 2

J8	J4	J7	J1	J3		J2
150	95	100	75	85		25
1	2	3	4	5	6	7

empty

(i) calculate Execution

(ii) which day is empty

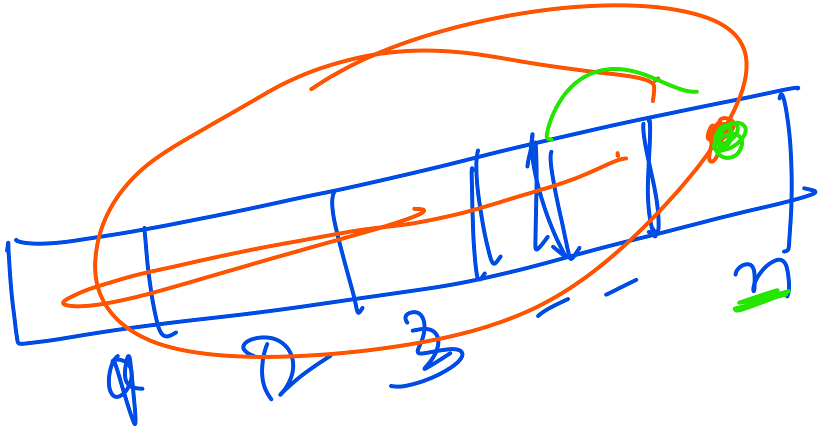
(iii) which Jobs left out

Time Complexity:

job	J_1	J_2	J_3	-- --	J_n
deadline	d_1	d_2	d_3	-- --	d_n
profit	p_1	p_2	p_3	-- --	p_n

① Sort the Jobs in non increasing order $= O(n\log n)$

②



$O(n^2)$

$1+2+3+4 - - n$

$n(n+1)/2 = O(n^2)$

$Tc = O(n^2) \Rightarrow$ worst case.

$Tc = O(n \log n) \Rightarrow$ Best and avg case